Corrigé — Exercice 1

$A = \left[\frac{3}{4}x^4 - \frac{2}{3}x^3 + \frac{1}{2}x^2 + 2x\right]_0^1 = \dfrac{31}{12}$.

$B : (x^2+2x)' = 2(x+1)$, donc $B = \left[\dfrac{-1}{2(x^2+2x)}\right]_1^2 = -\dfrac{1}{16} + \dfrac{1}{6} = \dfrac{5}{48}$.

$C : (2x^3+1)' = 6x^2$, donc $C = \left[\dfrac{1}{3}\sqrt{2x^3+1}\right]_0^1 = \dfrac{\sqrt{3}-1}{3}$.

$D : (x^2+6x+1)' = 2(x+3)$, donc $D = \left[\dfrac{(x^2+6x+1)^2}{4}\right]_0^1 = 16 - \dfrac{1}{4} = \dfrac{63}{4}$.

$E = \int_0^{\pi/3}\big(1 + \tan^2 x - 1\big)\,\mathrm{d}x = [\tan x - x]_0^{\pi/3} = \sqrt{3} - \dfrac{\pi}{3}$.

$F : (x^3+1)' = 3x^2$, donc $F = \left[\dfrac{2}{9}(x^3+1)^{3/2}\right]_0^2 = 6 - \dfrac{2}{9} = \dfrac{52}{9}$.

G : on pose $u = x+1$: $G = \int_1^4 \dfrac{u-1}{\sqrt{u}}\,\mathrm{d}u = \left[\frac{2}{3}u^{3/2} - 2u^{1/2}\right]_1^4 = \dfrac{4}{3} + \dfrac{4}{3} = \dfrac{8}{3}$.

$H = \left[\dfrac{(\ln x)^2}{2}\right]_1^{\mathrm{e}} = \dfrac{1}{2}$.